Longitudinal Detection of Contributor Teams in Wikipedia

Clustering Algorithms in Temporal Graphs

Master 2 Research Internship

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This Master 2 internship aims to develop and evaluate algorithms for longitudinal detection of contributor teams on Wikipedia. The challenge is twofold: on the technical level, it involves designing temporal graph clustering methods capable of tracking the coherent evolution of communities over time; on the application level, these methods must enable analysis of collaborative dynamics on Wikipedia by integrating organizational metrics from management science. The algorithms will be evaluated on academic benchmarks and then applied to real Wikipedia projects to characterize the formation, evolution, and dissolution of editorial teams. This work will contribute both to dynamic community detection methods and to understanding large-scale collaborative collectives.

Keywords: Temporal Networks, Dynamic Community Detection, Wikipedia, Collaborative Teams, Organizational Metrics

General Context

This internship is part of the PEPR eNSEMBLE - targeted project **CONGRATS** (COllectives for kNowledge pRoduction mAnagement That Scale), which studies the collaboration of large-scale open communities using digital platforms to create knowledge and value.

Wikipedia, with over 50 active language versions and millions of contributors, constitutes a privileged observation field for understanding how contributor teams form, evolve, and coordinate within editorial projects. These collectives exhibit complex dynamics: teams form around specific themes, reorganize (merge, split), welcome new members and see others leave. Analyzing these team dynamics is essential for understanding massive online collaboration mechanisms, evaluating the organizational viability of these collectives, and ultimately improving the quality of collaborative production.

The methods developed in this internship will be directly transferable to the analysis of large collectives on various online platforms (social networks, open-source development, collaborative forums), but the Wikipedia case study will serve as the primary application field.

Scientific Problem

The analysis of online collaborative communities faces two major difficulties that limit the scope of current approaches.

First difficulty: lack of longitudinal coherence. The majority of dynamic community detection methods [4] adopt a two-step strategy: community detection at each time t, followed by *a posteriori* matching of results to reconstruct temporal trajectories. This approach has significant limitations for analyzing platforms like Wikipedia, where activity is inherently fluctuating across projects and periods. The division into temporal windows followed by matching generates noise and a lack of coherence in team tracking: the same team can be artificially fragmented or merged depending on activity variations. Recent work proposes algorithms capable of directly detecting longitudinal communities, considering temporal evolution from the detection phase [2, 6]. However, the properties and limitations of these algorithms remain poorly studied, particularly in real application cases.

Second difficulty: absence of organizational metrics. Detection algorithms rely almost exclusively on topological properties (density, modularity) without accounting for organizational relevance criteria of a team. Yet, management science research has identified critical dimensions for characterizing team viability and effectiveness: internal cohesion [5], skill diversity, core stability, peripheral renewal, or the capacity to integrate new members [3, 1]. These dimensions, essential for evaluating whether a detected

group constitutes a true ”team” in the organizational sense, are currently not considered in detection or evaluation processes.

Given these observations, this internship aims to contribute to the development of longitudinal detection tools that meet the specific needs of collaborative collective analysis.

Objectives

Evaluating Longitudinal Detection Algorithms

The internship will build upon two recent longitudinal approaches that directly address the first identified challenge. The approach by Boudebza et al. [2] proposes an evolutionary detection method that integrates temporal continuity constraints, while the work by Trower et al. [6] extends spectral clustering to the spatio-temporal domain by explicitly coupling successive instances of the same node across time.

The internship will include an implementation phase for these two algorithms, followed by comparative evaluation on academic benchmarks (particularly DynBenchmark). The evaluation metrics will assess both partition quality and temporal coherence of detected trajectories. This comparison will quantify their respective strengths and identify the most effective mechanisms for tracking team transformations (mergers, splits, growth, decline).

Depending on the results, other longitudinal approaches may be explored before applying the methods to real Wikipedia data. To measure the actual contribution of temporal integration, these algorithms will also be compared against baseline methods that detect communities independently at each time step before attempting to match them.

Application Challenges and Integration of Organizational Metrics

Beyond algorithmic performance, the objective is to evaluate to what extent detected communities correspond to relevant teams from an organizational perspective. A first step will consist of characterizing detected communities using metrics from management research: internal cohesion [5], collective viability [3], collaborative effectiveness [3, 1], core-periphery structure. This qualitative evaluation will allow comparing algorithms not only on topological criteria, but also on their ability to identify groups that make sense for managerial analysis.

Applying algorithms to real Wikipedia projects (for example, thematic WikiProjects) will illustrate the diversity of results and understand how algorithmic choices influence the vision of collaborative dynamics.

Scientific Valorization and Dissemination

An essential objective of the internship is producing a scientific article presenting comparative results of evaluated algorithms, targeting an international conference in the field of complex networks. Simultaneously, all developed code will be made available to the scientific community under an open source license, in the spirit of reproducibility and sharing that characterizes the CONGRATS project. This dual valorization — publication and code dissemination — is essential for making a lasting contribution to research on dynamic communities and facilitating the adoption of these methods by other researchers.

Available Resources

- **Data**: Data from Wikipedia projects (edit history on articles and discussion pages) will be provided by the team
- **Benchmark**: DynBenchmark, Python library developed by the DECIDE team for evaluating dynamic community detection algorithms, including network generation with ground-truth and metric computation
- **Development Tools**: Python, NetworkX, temporal graph analysis libraries developed by the team

Desired Profile

We are seeking a Master 2 Research student with the following skills:

- Graph algorithmics
- Proficiency in Python (NetworkX, SciPy, pandas)
- Autonomy and organizational capacity

Internship Conditions

- **Duration**: 5 to 6 months
- **Period**: Starting March/April 2026
- **Location**: IMT Atlantique, Data Science Department, Brest Campus
- **Supervision**: Laurent Brisson (Director of Studies), Cécile Bothorel (Professor) - Lab-STICC Laboratory
- **Compensation**: According to current regulations
- **Prospects**: Possibility of continuing with a PhD (subject to obtaining funding)

Application Procedures

Deadline: January 31, 2026

Application file to be sent to laurent.brisson@imt-atlantique.fr **and** cecile.bothorel@imt-atlantique.fr including:

- Detailed CV
- Grade transcripts for L3, M1, M2
- Cover letter explaining your interest in the subject
- Names and contacts of two references

Bibliographic References

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